

Multimodal Text-Guided Artistic Photo Generation using Generative AI

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DA627 – Multimodal Generative AI
Project Report

1 Introduction

In recent years, Generative AI has made it much easier to create and edit digital content. Earlier, applying artistic styles to images required manual effort and good design skills. Now, with deep learning models, this process can be automated.

In this project, I have developed a system that generates artistic images using both photos and text prompts. The idea is to combine image processing and natural language understanding so that users can easily transform their images either by giving a reference style or simply describing it in words.

The project is divided into two phases. In the first phase, I used neural style transfer to apply styles from one image to another. In the second phase, I extended this by using models like CLIP and Stable Diffusion, which allow generating images based on text descriptions.

I also built a web interface where users can upload images, enter prompts, and see the results. The system shows outputs in real time, making it interactive and easy to use.

Overall, this project helped me understand how different AI models can work together to create artistic images while keeping the original structure of the photo.

2 Objectives

The primary objectives of this project are as follows:

- Develop a multimodal generative AI framework that combines image inputs and text prompts for artistic image generation.
- Implement a feature extraction pipeline using deep convolutional neural networks to capture structural information from photographs.
- Integrate text encoding mechanisms to interpret user-provided artistic style descriptions.
- Generate stylized images that preserve the structural content of the original photograph while applying artistic transformations.
- Design a web-based interface that allows users to upload photographs, provide text prompts, and download generated outputs.
- Evaluate the effectiveness of multimodal image generation techniques for creative media applications.

3 Dataset

For training and evaluating the system, publicly available datasets with image-text pairs and artistic images are used.

3.1 Flickr30k Dataset

The Flickr30k dataset contains around 30,000 images, each with multiple textual descriptions. It is commonly used in tasks like image captioning and multimodal learning. In this project, it helps the model understand the relationship between images and natural language.

3.2 WikiArt Dataset

The WikiArt dataset includes thousands of paintings from different artistic styles such as impressionism, cubism, surrealism, and abstract art. This dataset is useful for learning artistic patterns and styles that can be applied to images.

3.3 LAION Subset

A subset of the LAION dataset is used, which contains large-scale image-text pairs. It is mainly used for understanding multimodal alignment between visual and textual data and supports the text-guided image generation process.

4 Models Used in the System

This project uses different deep learning models for image processing, multimodal understanding, and image generation. The system is divided into two phases, where each phase uses different models.

4.1 Neural Style Transfer (VGG19)

In Phase 1, Neural Style Transfer is used to convert a normal image into an artistic image using a reference style image. This method is based on a pre-trained convolutional neural network called VGG19.

The model extracts two types of features:

- **Content Features:** These represent the structure of the image such as shapes and objects.
- **Style Features:** These represent textures, colors, and artistic patterns.

Style features are captured using Gram matrices, which help in understanding the texture patterns of the reference image. The final output image is generated by minimizing a loss function that combines:

- Content Loss (to preserve structure)
- Style Loss (to apply artistic style)
- Total Variation Loss (to smooth the image)

An optimization algorithm called L-BFGS is used to update the generated image iteratively until a good balance between content and style is achieved.

4.2 CLIP Model (Multimodal Understanding)

In Phase 2, the CLIP model is used to understand the relationship between images and text. CLIP (Contrastive Language-Image Pretraining) is trained on a large dataset of image-text pairs.

It consists of two main parts:

- Image Encoder (processes images)
- Text Encoder (processes text prompts)

Both encoders convert inputs into a shared feature space. This allows the model to measure how well a text description matches an image. In this project, CLIP helps guide the generation process so that the output image follows the given text prompt.

4.3 Stable Diffusion (Image Generation)

Stable Diffusion is used for generating images based on text prompts. It is a type of diffusion model that works by gradually removing noise from an image.

The process works as follows:

- Start with a noisy version of the image
- Use the model to remove noise step by step
- Guide the process using text embeddings

Stable Diffusion uses a Variational Autoencoder (VAE) to convert images into a smaller latent space, which makes the process faster and more efficient.

It also uses a technique called **classifier-free guidance**, which controls how strongly the generated image follows the text prompt. A higher guidance value results in outputs that match the prompt more closely.

4.4 Overall Model Integration

The system combines these models to create a complete pipeline:

- Phase 1: Uses VGG19 for style transfer
- Phase 2: Uses CLIP for understanding text and Stable Diffusion for generation

5 Methodology

The proposed system follows a multimodal generative approach where both image and text inputs are used to generate artistic outputs. The complete process is divided into multiple stages, each handling a specific task in the pipeline.

5.1 Image Input

In this step, the user uploads an image that acts as the base content image. The system preprocesses the image by resizing it and normalizing pixel values so that it can be used by deep learning models without any issues.

5.2 Text Prompt Encoding

The user provides a text prompt describing the desired artistic style. Some example prompts include:

- “Convert this photo into a watercolor painting”
- “Apply Van Gogh painting style”
- “Cinematic photography style with dramatic lighting”

This text is processed using a transformer-based encoder to convert it into meaningful numerical representations (embeddings), which can be used by the model.

5.3 Image Feature Extraction

The system extracts important visual features from the input image using deep learning models such as VGG19. These features represent the structure, objects, and textures present in the image.

5.4 Multimodal Alignment

In this stage, both image features and text embeddings are aligned using the CLIP model. CLIP maps both inputs into a common feature space, allowing the system to understand how well the text description matches the image.

5.5 Generative Image Transformation

The final image is generated using two approaches:

- Neural Style Transfer (Phase 1) for image-to-image transformation
- Stable Diffusion (Phase 2) for text-guided generation

The system combines the structural details of the original image with the style information from either a reference image or text prompt to produce the final output.

5.6 Output Generation

The generated image is displayed through a web interface. The system shows results in real time along with processing logs, and users can download the generated images for further use.

6 Technologies Used

The implementation will utilize several modern machine learning and software development tools.

- Python for overall development
- TensorFlow and PyTorch for deep learning models
- CLIP and Stable Diffusion for multimodal generation
- OpenCV and PIL for image processing
- Flask for backend development
- HTML, CSS, and JavaScript for frontend interface

7 Results and Performance

The system was tested on different images and prompts to evaluate its performance in generating artistic outputs.

7.1 Phase 1 Results

In Phase 1, the neural style transfer model successfully applied artistic styles from reference images while preserving the structure of the original image.

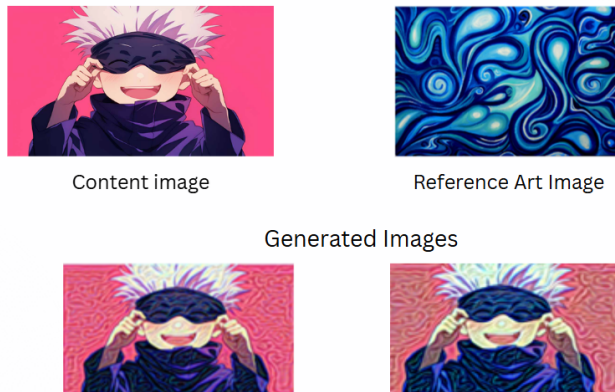


Figure 1: Image to Image Transformation

7.2 Phase 2 Results

In Phase 2, the system generated images based on text prompts using Stable Diffusion. Multiple variations provided different creative outputs.

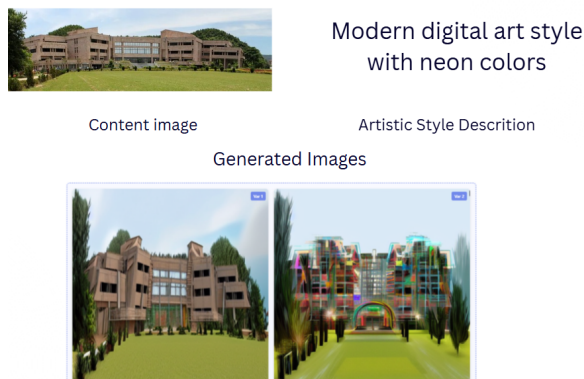


Figure 2: Images based on text prompt

7.3 Performance Analysis

The performance of the system depends on the model and hardware used.

- Phase 1 processing time: 30–120 seconds per image
- Phase 2 processing time: 20–60 seconds (GPU)
- Supports both CPU and GPU execution

Higher quality results require more processing time, especially when increasing diffusion steps or image resolution.

8 User Interface

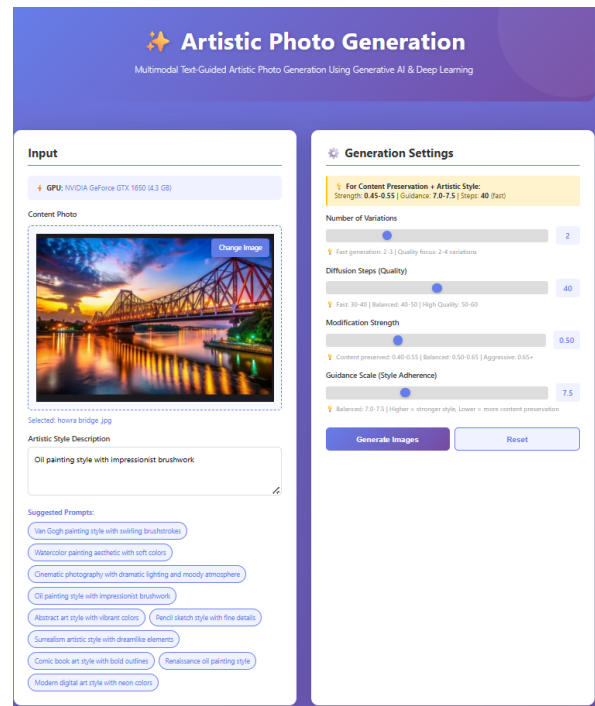


Figure 3: Main Dashboard

The web interface provides real-time feedback during processing.

- Live terminal logs during execution

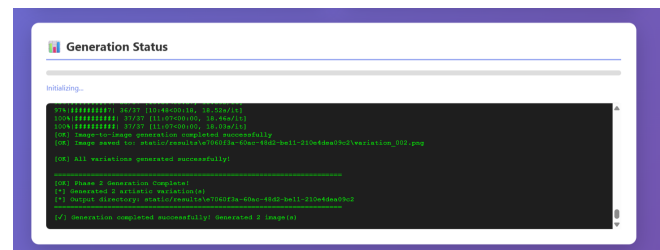


Figure 4: Terminal Output

- Easy download of generated outputs

Generated Results

Generated 2 artistic variation(s) for prompt: Oil painting style with impressionist brushwork

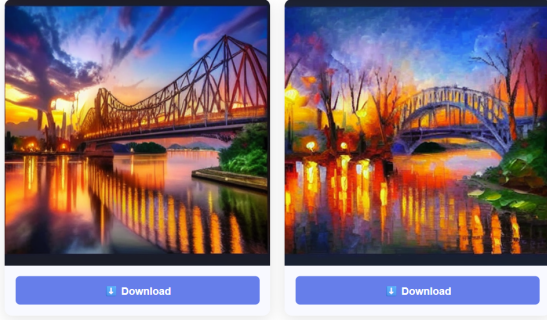


Figure 5: Generated Results

9 Relevance and Future Impact

This project helped me strengthen my understanding of multimodal generative AI by combining concepts from computer vision, natural language processing, and deep learning. Through the implementation of text-guided artistic image generation, I have gained practical experience with modern generative models such as diffusion models and multimodal encoders. In the short term, this project will enhance my skills in model development, dataset handling, and system integration for AI applications. In the long term, the knowledge gained from this work can be useful in fields such as creative AI systems, automated image editing, digital content generation, and intelligent media applications, where multimodal generative models are becoming increasingly important.

10 Future Scope

The proposed system can be extended in several directions. Future work may include supporting real-time image generation, expanding the range of artistic styles, and integrating additional modalities such as audio descriptions or sketch inputs. The system can also be extended for applications in digital art creation, social media content generation, and automated image editing tools.

11 Conclusion

In this project, a multimodal generative AI system was developed to generate artistic images using both image inputs and text prompts. The system successfully combines traditional image-based techniques with modern text-guided generation methods.

In Phase 1, neural style transfer was implemented to apply artistic styles from reference images while preserving the structure of the original image. In Phase 2, the system was extended using CLIP and Stable Diffusion to generate images based on textual descriptions, providing more flexibility to users.

The integration of these models into a single pipeline, along with a web-based interface, made the system interactive and easy to use. Users can upload images, provide prompts, and generate results in real time.

Overall, the project demonstrates how multimodal AI can be effectively used for creative applications such as artistic image generation. It also helped in understanding the practical implementation of deep learning models and their integration into real-world systems.

References

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